

Form Filler

Word Form Extractor & Generator

A single-file, browser-based tool that reads a blank Word (.docx) questionnaire, turns every question into a clean input field, and writes the answers back into the original document — producing a completed form with the layout intact, without the file ever leaving the user's computer.

01

It reads the form

Every question in the document's table is detected automatically.

02

You answer

Type into a clean field for each question — no scrolling through the doc.

03

Export filled .docx

Answers are written into the original layout and downloaded as Word.

DOC REF	VERSION	DATE	STATUS	PLATFORM
FF-2026-001	1.0	July 15, 2026	Released	Web · Client-side

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Executive Summary

Project Name

Form Filler — Word Form Extractor & Generator

Objective

To automate the process of reading blank Word (.docx) questionnaires, presenting every question as a structured on-screen input field, and generating the completed Word document automatically — while minimizing manual effort, reducing formatting errors, and improving turnaround time.

The solution aims to:

- Eliminate repetitive manual form-filling inside Word tables
- Preserve the original document layout and formatting on export
- Reduce data-entry and formatting errors
- Standardize recurring answers using editable presets
- Keep documents fully private by processing everything in the browser

What is the Form Filler?

The Form Filler is a browser-based tool that reads a blank Word (.docx) form, automatically extracts every question from the document's table structure, and provides a clean digital form to answer them. Once completed, the tool writes the answers back into the original document layout and exports a filled Word file — without the document ever leaving the user's computer.

Instead of manually scrolling through a Word document and typing into cramped table cells, the system automatically:

- Reads the document's internal XML structure
- Detects every question, guidance note, and answer cell
- Presents each question with the right input control (text, multi-line, Yes/No)
- Pre-fills standard answers where configured
- Tracks completion progress in real time
- Writes answers back into the original cells and exports the filled file

EXAMPLE

WITHOUT THE FORM FILLER

An executive receives a 30-question vendor compliance questionnaire in Word, scrolls through the table, clicks into each answer cell, retypes standard company details, and manually repairs formatting that breaks along the way.

WITH THE FORM FILLER

The executive drops the file into the tool, answers each question in a clean form — with standard details already pre-filled — and downloads the completed document in minutes, formatting intact.

SECTION 03

Why is the Form Filler Necessary?

Current Challenges

Organizations and individuals working with Word-based forms routinely face:

Manual effort

Filling table-based forms means constant scrolling, clicking into small cells, and switching between reading questions and typing answers.

Formatting errors

Broken table layouts, inconsistent fonts, answers typed into the wrong cells, and accidental deletion of question text.

Repetitive data entry

Standard details — company name, registration numbers, business category — are retyped on every new form.

No progress visibility

In a long document it is hard to know how many questions remain unanswered before the form can be submitted.

Privacy risk

Online form-filling and converter services require uploading confidential business documents to third-party servers.

SECTION 04

Business Problem Statement

Existing Situation

Business teams frequently complete Word-based forms for:

- Vendor onboarding and registration
- Compliance and KYC questionnaires
- Client due-diligence requests
- Audit and certification checklists

- Internal declarations and HR forms

These forms are currently filled manually inside Microsoft Word, which consumes valuable operational time and introduces avoidable errors.

Business Impact

- Reduced employee productivity
- Inconsistent answers across submissions
- Formatting rework before documents can be sent
- Delayed responses to clients and regulators
- Risk of confidential documents being uploaded to unknown online tools

Proposed Solution

Implement a lightweight tool that reads any table-based Word form, converts it into a structured digital form, and regenerates the completed document automatically — without the file ever leaving the user's computer.

SECTION 05

Technology Stack Evaluation & Solution Recommendation

OPTION 1

Single-File Browser Application

RECOMMENDED

TECHNOLOGY STACK

HTML / CSS / JavaScript

JSZip

DOMParser / XMLSerializer

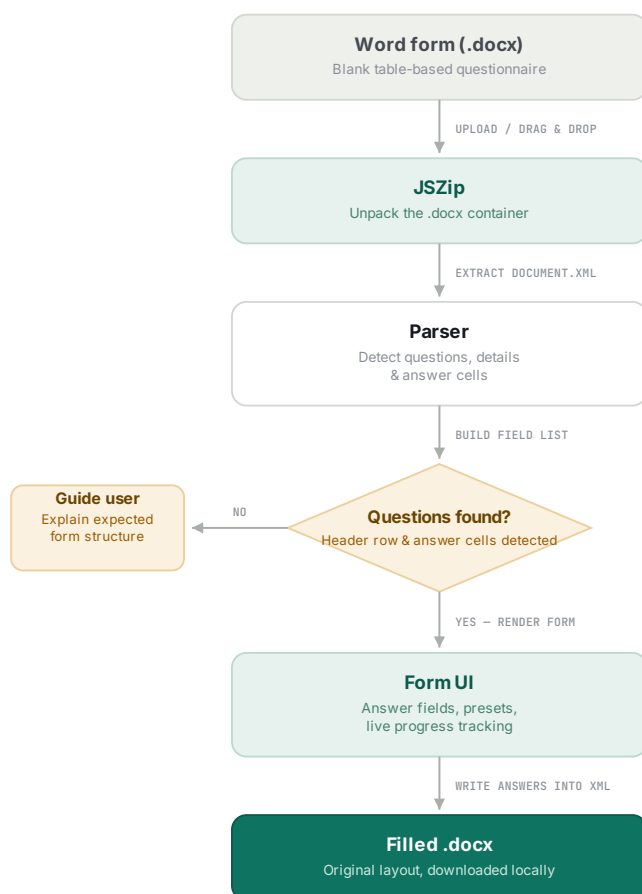
File & Blob APIs

ADVANTAGES

- Zero infrastructure — runs entirely in the browser
- 100% private: the document never leaves the user's machine
- No installation; works in any modern browser
- Free to host (e.g., GitHub Pages) and effortless to maintain
- Preset answers standardize recurring responses
- Preserves the original document formatting on export

ARCHITECTURE FLOW

WORD FORM → JSZIP → PARSER → FORM UI → FILLED .DOCX



LIMITATIONS

- Supports table-based forms only
- Works with .docx files only (not .doc or PDF)
- Not designed for multi-user collaboration or centralized records

BEST FIT

Individuals and teams needing fast, private form completion with zero setup or IT involvement.

OPTION 2

Custom Web Application

TECHNOLOGY STACK

React

Node.js

PostgreSQL

Server-side document engine

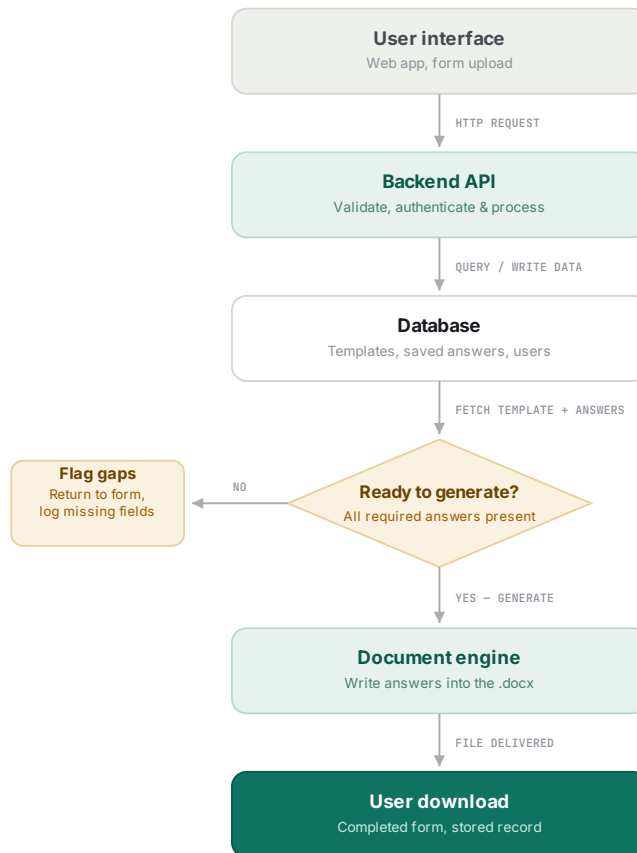
ADVANTAGES

- Centralized template and answer library
- Multi-user access with roles and permissions
- Submission history, audit trails, and analytics
- Organization-wide standardization of answers

- Integration with internal systems (CRM, ERP)

ARCHITECTURE FLOW

USER INTERFACE → BACKEND API → DATABASE → DOCUMENT ENGINE → DOWNLOAD



LIMITATIONS

- Higher development cost and longer implementation timeline
- Requires server hosting and ongoing maintenance
- Documents are processed on a server, so privacy controls are required

BEST FIT

Organizations managing large volumes of forms across many users and departments.

OPTION 3 Cloud-Native Enterprise Solution

TECHNOLOGY STACK

AWS Lambda

Amazon S3

DynamoDB

API Gateway / EventBridge

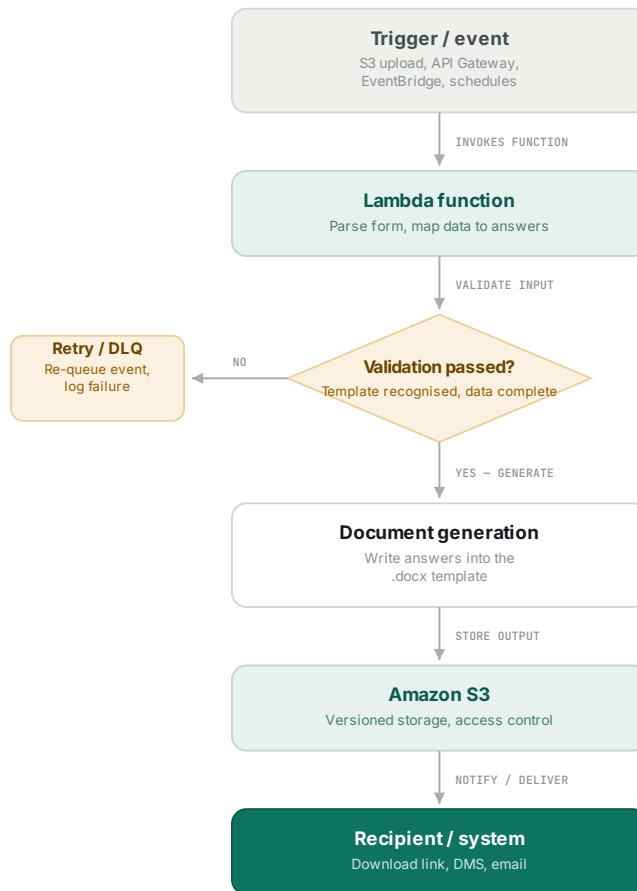
ADVANTAGES

- Highly scalable, event-driven processing
- Forms can be filled automatically from existing data sources
- Enterprise-grade reliability and monitoring

- Cost-efficient at very high volumes
- Integrates with document management systems

ARCHITECTURE FLOW

TRIGGER / EVENT → LAMBDA → DOCUMENT GENERATION → S3 → RECIPIENT



LIMITATIONS

- Higher technical complexity
- Requires cloud expertise
- Longer setup time

BEST FIT

Large organizations processing high volumes of forms programmatically across systems.

5.4 Solution Comparison

Criteria	Single-File Browser App RECOMMENDED	Custom Application	AWS Cloud Solution
Implementation Cost	Very Low	Medium	High
Maintenance Effort	Very Low	Medium	Medium
Scalability	Medium	High	Very High
Infrastructure Requirement	None	Required	Required
Data Privacy	Very High (fully local)	Medium	Medium
Customization Capability	Medium	High	Very High
Technical Complexity	Low	Medium	High
Operational Overhead	None	Medium	High
Business User Accessibility	High	Medium	Low
Enterprise Readiness	Medium	High	Very High

FINAL RECOMMENDATION

Single-File Browser Application — HTML + JavaScript + JSZip

Justification

After evaluating multiple implementation approaches, the single-file browser-based solution is recommended because:

- It fully addresses the current requirement of automating repetitive Word form filling.
- It requires no infrastructure, hosting, or installation.
- It keeps every document 100% private on the user's machine.
- It is the fastest option to implement, distribute, and maintain.
- Preset answers ensure consistent, standardized responses across forms.
- It supports future migration to a centralized web application if requirements expand.

SECTION 06

Conclusion

- The Form Filler Project streamlines and automates the completion of table-based Word forms. By replacing manual editing inside Word with a structured digital form, the solution improves

productivity, reduces formatting errors, standardizes recurring answers, and ensures documents remain private.

- After evaluating multiple technology approaches, including a Custom Web Application and a Cloud-Native Enterprise Solution, the single-file browser application has been identified as the most suitable option for the current requirement. It offers a zero-cost, zero-infrastructure, and easy-to-maintain foundation while keeping every document fully local.
- The proposed solution aligns with immediate needs and provides the flexibility to evolve into centralized or cloud-based architectures as business requirements grow in the future.

KEY OUTCOMES EXPECTED

- Reduced manual effort in completing Word forms
- Improved accuracy and formatting consistency
- Faster turnaround on questionnaires and compliance requests
- Standardized answers through reusable presets
- Complete document privacy — fully client-side
- Scalable path toward centralized automation

LIVE DEMO

imaginative-marshmallow-a28cfc.netlify.app

SOURCE REPOSITORY

github.com/Sarvesh-c07/Form-Filler-Extractor_-_Generator